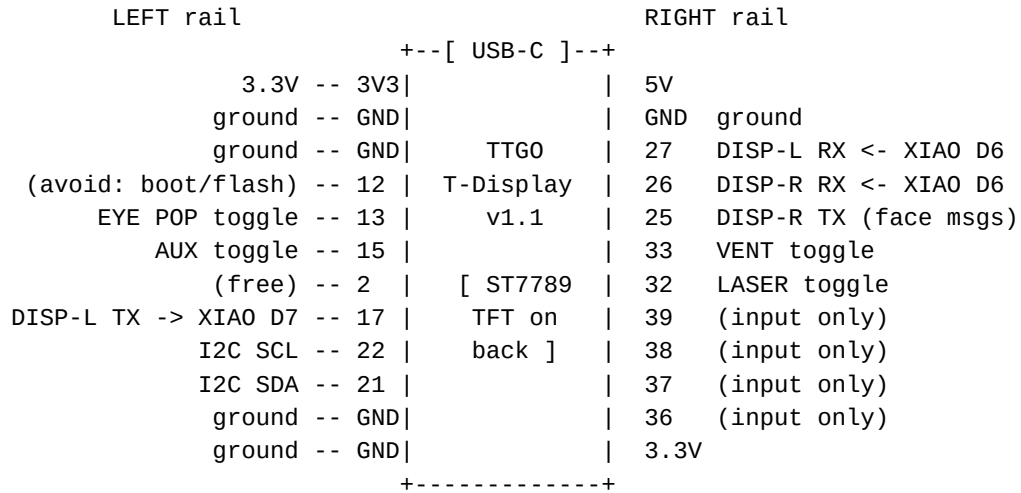


# j4\_controller -- pin diagram

TTGO T-Display v1.1, rails visible, USB-C at the TOP (as mounted)



on-board (no header pin): GPIO34 = battery sense, BOOT = GPIO0,  
 GPIO35 = screen-cycle button, TFT on 4/5/16/18/19/23  
 I2C bus: ADS\_01 @ 0x48, ADS\_02 @ 0x49, ADS\_03 @ 0x4A, ADS\_04 @ 0x4B,  
 PCF8574 keypad\_left @ 0x21, keypad\_right @ 0x20  
 toggles: INPUT\_PULLUP, switch closes to GND (ON = LOW)

| GPIO | Function                                                            |
|------|---------------------------------------------------------------------|
| 21   | I2C SDA (ADS1115 x4, PCF8574 keypads x2)                            |
| 22   | I2C SCL                                                             |
| 17   | Serial1 TX to j4_display_left (XIAO D7)                             |
| 27   | Serial1 RX from j4_display_left (XIAO D6)                           |
| 25   | Serial2 TX to j4_display_right (XIAO D7, face-preset messages)      |
| 26   | Serial2 RX from j4_display_right (XIAO D6, pot feed)                |
| 32   | LASER toggle (INPUT_PULLUP, switch closes to GND)                   |
| 33   | VENT toggle (INPUT_PULLUP, switch closes to GND)                    |
| 13   | EYE POP toggle (INPUT_PULLUP, switch closes to GND)                 |
| 15   | AUX toggle, next to the right joystick (INPUT_PULLUP, unassigned)   |
| 34   | Battery voltage sense (ADC input, read-only)                        |
| 35   | Screen-cycle button (TTGO built-in; data / MAC / connection status) |

# j4\_controller -- ADS1115 pin diagrams

pot wiper to the A-pin; outer legs to 3.3V and GND

```
ADS_01 @ 0x48 (ADDR -> GND)                both joysticks
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | GND = address 0x48
| ALRT | not connected
| A0 | neck joystick X wiper      -> neck L/R differential mix
| A1 | neck joystick Y (jaw) wiper -> neck L/R base height
| A2 | eyes joystick X wiper      -> eyes pan servo (PCA9685 ch 3), -128..0..128
| A3 | eyes joystick Y wiper      -> eyes tilt servo (PCA9685 ch 4), -128..0..128
+-----+

ADS_02 @ 0x49 (ADDR -> VDD)                four face pots, left bank
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | VDD = address 0x49
| ALRT | not connected
| A0 | Eyebrow L pot wiper        -> PCA9685 ch 6
| A1 | Eyebrow R pot wiper        -> PCA9685 ch 7
| A2 | Basket Eyebrow L pot wiper -> PCA9685 ch 8
| A3 | Basket Eyebrow R pot wiper -> PCA9685 ch 9
+-----+

ADS_03 @ 0x4A (ADDR -> SDA)                three face pots, right bank
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SDA = address 0x4A
| ALRT | not connected
| A0 | Nose pot wiper (up/down)   -> PCA9685 ch 10
| A1 | FAULTY channel, nothing connected, left unread in firmware
| A2 | Bottom Eyelid L pot wiper  -> PCA9685 ch 12
| A3 | Bottom Eyelid R pot wiper  -> PCA9685 ch 13
+-----+

ADS_04 @ 0x4B (ADDR -> SCL)                neck-pivot + faders + Nose Basket
+-----+
| VDD | 3.3V
| GND | GND
| SCL | TTGO GPIO 22 (I2C SCL)
| SDA | TTGO GPIO 21 (I2C SDA)
| ADDR | SCL = address 0x4B
| ALRT | not connected
| A0 | NECK PIV pot -- DISABLED, read for display only (to be repurposed)
| A1 | fader_left wiper           -> neck pivot, sole source (-2048..0..2048)
| A2 | fader_right wiper          -> iris (doubles j4_display_right IRIS), 0-255
| A3 | Nose Basket pot wiper      -> PCA9685 ch 11 (single source)
+-----+
```